

Power SiC DMOSFET Model Accounting for Nonuniform Current Distribution in JFET Region

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Abstract—The main goal of this paper is development of a new circuit-based silicon carbide (SiC) DMOSFET model which physically represents the mechanism of current saturation in power SiC DMOSFET. Finite-element simulations show that current saturation for a typical device geometry is due to 2-D carrier distribution effects in the JFET region caused by current spreading from the channel to the JFET region. For high drain-source voltages, most of the voltage drop occurs in the current spreading region located in the JFET region close to the channel. A new model is proposed that represents the nonuniform current distribution in the JFET region using a nonlinear voltage source and a resistance network. Advantages of the proposed model are that a single set of equations describes operation in both the linear and saturation regions, and that it provides a more physical description of MOSFET operation.

Index Terms—Current spreading region, DMOSFET, nonuniform current distribution, silicon carbide (SiC).

NOMENCLATURE

β	Constant for saturation velocity.
ϵ_{SiC}	Dielectric constant of silicon carbide.
μ_{CH0}	Low-field mobility in inversion layer.
$\mu_{n\text{-drift}}$	Carrier mobility in drift region.
μ_{CH}	field-dependent carrier mobility at the end of the channel.
μ_A	Accumulation layer mobility.
μ_{JFET}	JFET region mobility.
a	Half of width of channel region.
a_i	Ratio of intercell area to total die area.
A	Total die area.
C_{GD}	Capacitance between gate and drain.
C_{OX}	Oxide capacitance per unit area.
c_{OX}	Oxide capacitance.

E_k	Peak electric field at the end of channel under forward bias conditions.
I_D	Drain current.
L_{CH}	Length of channel.
L_{DRIFT}	Length of drift region.
L_{JFET}	Length of JFET region.
L_R	Distance between adjacent nodes in JFET region.
N_{DRIFT}	Doping concentration of drift region.
N_{JFET}	Doping concentration of JFET region.
q	Fundamental electronic charge.
R_{ai}	Accumulation layer resistance.
R_{DRIFT}	Drift region resistance.
R_{J_net}	Resistance of JFET region network.
v_{sat}	Carrier saturation velocity in silicon carbide.
v_1	Electron velocity at the end of channel.
V_{bi}	Built-in voltage in gate source junction.
V_{CH}	Channel voltage.
V_{DS}	Voltage applied to drain and source.
V_{GS}	Voltage applied to gate and source.
V_{GD}	Voltage applied to gate and drain.
V_i	Voltage at the node of corresponding i resistor.
V_{JFET}	Total voltage in JFET region.
V_T	Threshold voltage.
W'	Length of the depletion region under the gate oxide.
W_{DRIFT}	Width of drift region.
W_{DS}	Width of depletion region between drain and source.
W_{GS}	Width of depletion region between gate and source.
W_{JFET}	Depletion layer length of the depletion region under the gate which only begins to grow when $V_{\text{DS}} = V_{\text{GS}}$.
Z	Length of device in vertical-to-paper dimension.

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I. INTRODUCTION

FOR SWITCHING converter applications at less than 200 V, the silicon power MOSFET has become the device of choice due to its low on-state resistance and fast switching speed. When designed for higher operating voltages, the use of silicon MOSFETs becomes impractical due to the very high drift region resistance. For higher blocking voltages, insulated gate bipolar transistors (IGBTs) and gate turn-off thyristors (GTOs) are commonly used. However, these devices have relatively high on-state voltage drop and relatively slow switching speed. Excellent electrical properties of silicon carbide (SiC)

material make it a very attractive semiconductor material for power switching devices with capabilities that are superior to those of devices based on silicon technology. In particular, 4H-SiC MOSFET is one of the most promising candidates for high-speed and low-loss power switching applications [1]–[3].

Thanks to recent progress in SiC technology [4], [5], SiC MOSFETs are on the verge of commercialization. Since SiC MOSFETs are still under development, there is a need to create accurate models for SiC prototype devices so that engineers can explore through circuit-level simulation the advantages that the introduction of these devices can provide in their switching converter designs.

Device models can be divided in two major groups: 1) analytical models based on the finite-element solution of drift-diffusion carrier transport in two or three dimensions [6]–[8] and 2) circuit-oriented models which employ equation-based description of device behavior. Analytical models provide very high accuracy but require long simulation time and detailed information about device fabrication, while circuit-based models require much less time for simulation with acceptably accurate results using model parameters that can be extracted from experimental measurements. An additional advantage of circuit-oriented models is that they are compatible with circuit simulators and can be used to simulate an entire switching converter. While extensive research has been done to develop analytical models for SiC MOSFET [9]–[11], there are very few publications addressing the implementation of circuit-oriented models in simulators such as PSpice [12], [13].

The physics-based numerical model in [9], which accounts for incomplete ionization and the charging and discharging of interface states, is based on the drift-diffusion equations. This model could reveal the effects of surface phenomena, but it is quite complex and would require long simulation times. Moreover, the model is not suitable for implementation in a circuit simulator due to its complexity. The model in [13] is a simple SPICE behavioral model for SiC power DMOSFETs. The model is obtained by introducing specific modifications to the conventional level-1 Si power MOSFET model available in commercial SPICE simulators. The model in [12] is based on the latest version of the power MOSFET model utilized in the Hefner IGBT model [14], where the parameters of both the Si and SiC MOSFETs can be extracted using the IGBT Model Parameter ExtrAction Tool software [15]. Both these models need to extract several fitting factors from device experimental output characteristics to obtain good matching.

In this paper, a new circuit-based SiC DMOSFET model that physically represents the mechanism of current saturation in power SiC DMOSFET is developed in PSpice software based on finite-element simulation results. A big advantage of the model is that a single set of equations describes operation in both the linear and saturation regions. The model describes the nonuniform current distribution in the JFET region that causes current saturation, as shown by the finite-element simulations. Finally, the model is validated by comparing simulated results with experimental results both under static and dynamic conditions.

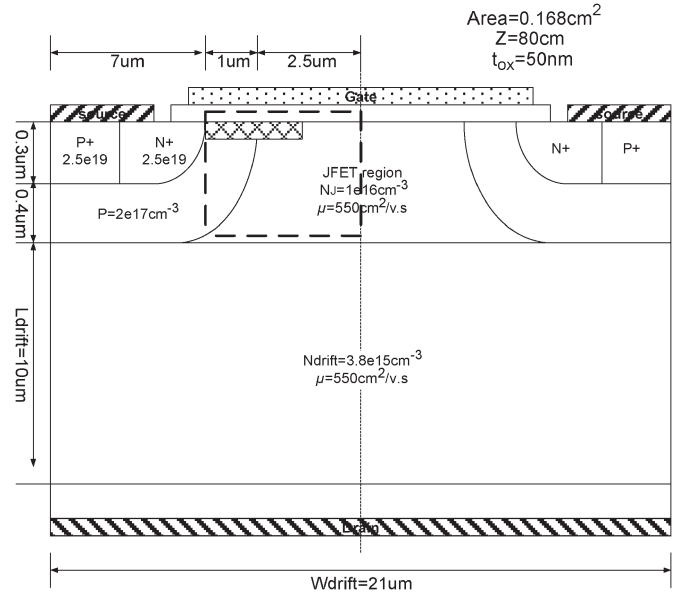


Fig. 1. Structure of power SiC DMOSFET.

II. FINITE-ELEMENT SIMULATION

Typical structure of a power SiC DMOSFET is shown in Fig. 1. The physical dimensions and doping concentrations are for a 1.2-kV device. It is a vertical device with a planar gate, the channel has a 1 μm length and the drift region has a 10 μm thickness. Applying a positive voltage to the gate larger than the threshold voltage, an inversion layer is created on top of the p-base region, forming an n-type conducting channel connecting the source to the JFET region. At the same time, an accumulation layer is formed under the gate oxide at the top of the JFET region, providing current spreading for the electron current flowing from the channel into the JFET region. For small values of drain-source voltage (linear or triode region), the device exhibits an approximately constant on-state resistance, which is determined by channel and drift region resistances. Ultimately, at a higher drain-source bias, the current saturates (saturation region). From standard analysis of power MOSFETs, it is known that there are at least three mechanisms that can contribute to current saturation in power MOSFET: channel pinch-off, carrier velocity saturation in high-electric-field regions, and the so-called parasitic JFET effect, i.e., the fact that the current does not flow uniformly in the JFET region but is constrained to flow in a narrow JFET channel. According to performed finite-element simulations, carrier velocity saturation at the interface between the channel and the JFET region is the main reason for current saturation for the SiC DMOSFET structure of Fig. 1. The JFET effect is also a significant contributing factor. The proposed physics-based circuit-oriented model captures these two effects.

Fig. 2 shows the forward I - V characteristic obtained from finite-element simulation of the device structure of Fig. 1 using Silvaco ATLAS [16]. Gate-source voltage is constant (15 V) for all drain-source voltages, ensuring that the device is fully on. Three values of drain-source voltage are considered: $V_{ds} = 5$ V, 15 V, and 30 V. For $V_{ds} = 5$ V the device operates in the linear region, and for $V_{ds} = 15$ V, 30 V, the device operates in the saturation region.

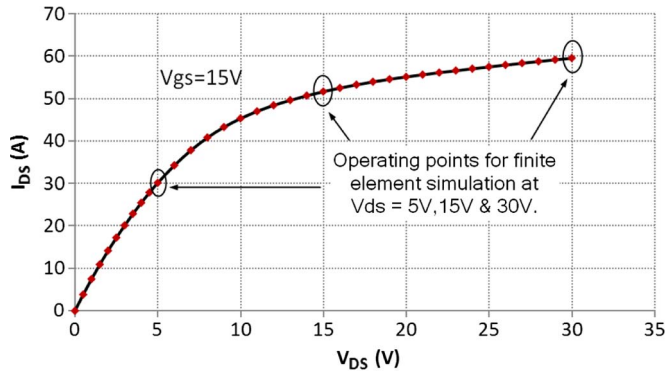


Fig. 2. I - V characteristic of DMOSFET for $V_{gs} = 15$ V obtained from finite-element simulation.

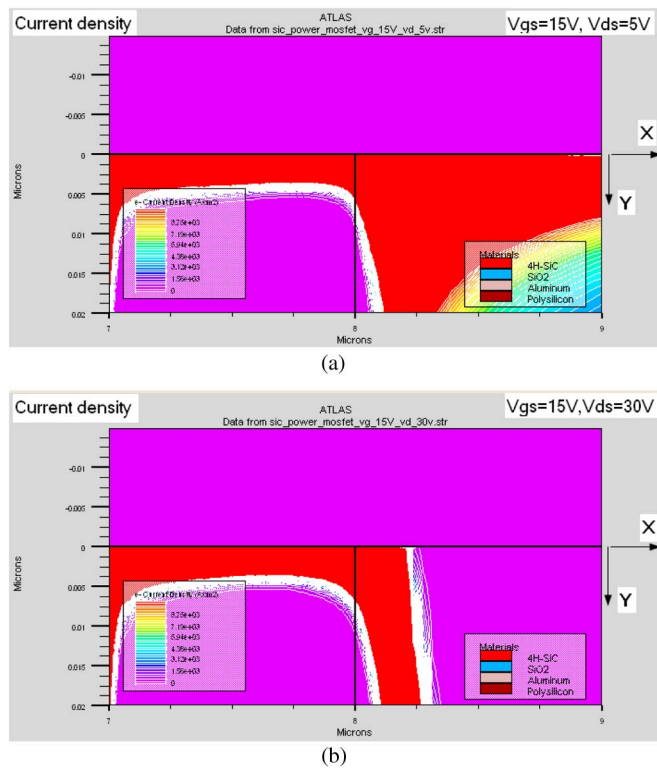


Fig. 3. Finite-element simulation results. Current density 2-D distribution along the channel for operation in the linear region, $V_{ds} = 5$ V (a) and in the saturation region, $V_{ds} = 30$ V (b). Gate-source bias is $V_{gs} = 15$ V.

Fig. 3(a) and (b) shows the current density distribution in a rectangular region encompassing the channel and the top portion of the JFET region for $V_{ds} = 5$ V and for $V_{ds} = 30$ V. This region extends for $2\text{ }\mu\text{m}$ in the x -direction and for $0.035\text{ }\mu\text{m}$ in the y -direction and is identified by the small hatched rectangular region in Fig. 1. Notice that the thickness and the shape of the channel are very similar for the two drain-source voltages and no significant pinch-off is observed even when the device is in deep saturation ($V_{ds} = 30$ V). This shows that channel pinch-off is not a significant contributor to current saturation. The biggest difference between Fig. 3(a) and (b) is that for operation in the saturation region, a depletion region forms in the JFET region, and the current flows along a narrow vertical path adjacent to the p -base.

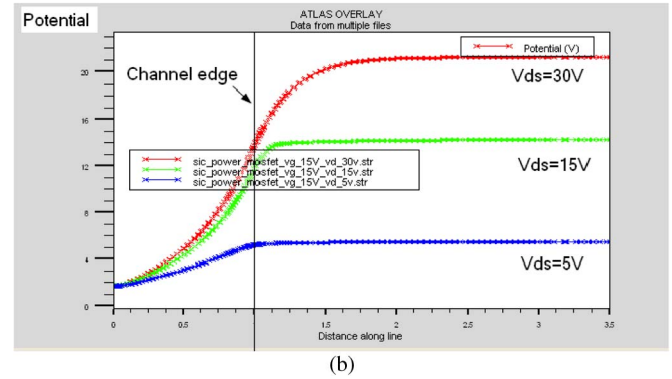
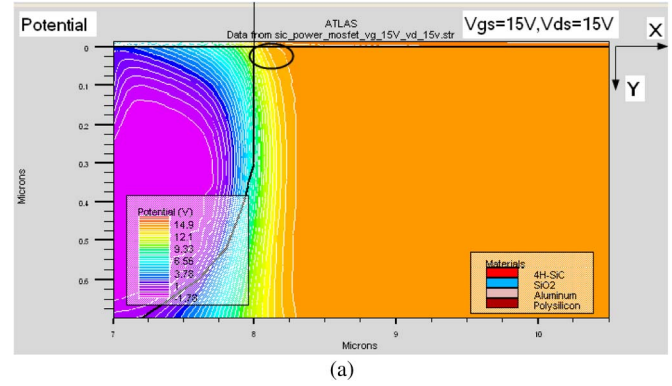


Fig. 4. Finite-element simulation results. Electric potential 2-D distribution for $V_{gs} = 15$ V, $V_{ds} = 15$ V (a) and electric potential along cutline for three drain-source voltages (b).

Fig. 4(a) shows 2-D finite-element simulation results for the electric potential in the upper part of the SiC DMOSFET (rectangular dashed region in Fig. 1). Fig. 4(b) shows the electric potential curve on a cutline along the channel (cutline: $X = 7\text{ }\mu\text{m}$ – $10.5\text{ }\mu\text{m}$, $Y = 1\text{ nm}$) for the three values of drain-source voltage of Fig. 2. Fig. 5(a) shows the 2-D current transport, and Fig. 5(b) shows the lateral (x -axis) electric field component along the same cutline.

From these simulation results, one can draw some conclusions on saturation mechanism in a typical power SiC DMOSFET. Pinch-off of the channel was not observed under any drain-source bias: even at $V_{ds} = 30$ V the channel retains its approximately rectangular shape and does not display channel pinch-off condition. Additional confirmation of this can be obtained by looking at Fig. 4(b), which shows that the potential drop in the channel does not change much (from 12 V to 14 V approximately) as V_{ds} increases from 15 V to 30 V. This shows that most of the voltage drop across the device in saturation does not occur in the channel, but somewhere else (in the JFET region). As the drain-source voltage increases, the lateral electric field shown in Fig. 5(b) becomes quite large at the end of the channel, where it reaches its peak value E_k when current goes to saturation regime. Detailed investigation of carrier transport in the channel region as predicted by the finite-element simulations reveals that carrier velocity saturation is responsible for the large electric field E_k and consequently is the main contributor causing current saturation in the device.

Traditionally, it is assumed that an accumulation layer is always formed under the gate oxide in the JFET region, and further analysis of current transport in power MOSFET is based

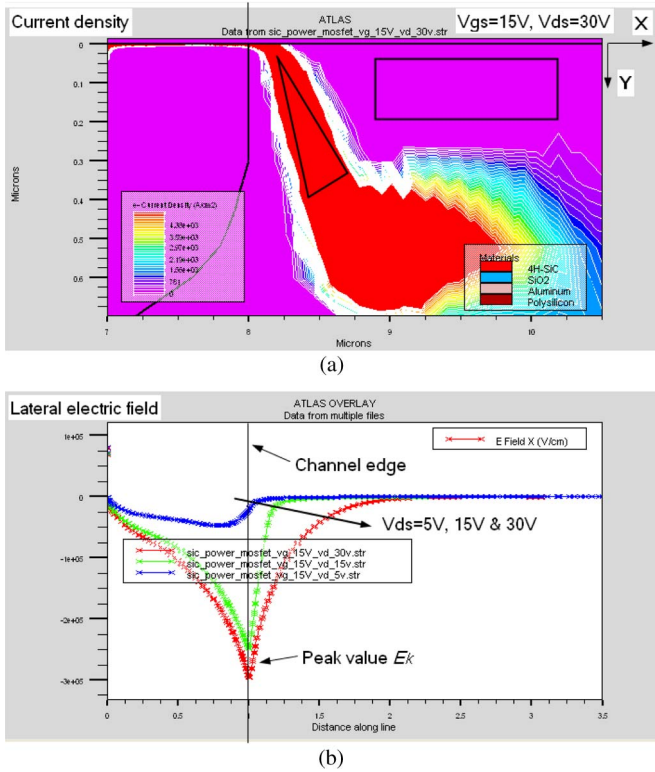


Fig. 5. Finite-element simulation results. Current density 2-D distribution (a) and lateral electric field along outline (b).

on this assumption. This accumulation layer helps spreading the electron current coming from the channel uniformly across the undepleted portion of the JFET region. As a result, the JFET region can be represented as a rectangular piece of semiconductor material whose width is modulated by change of depletion region width of p-base/n-JFET junction, with electron current flowing vertically from the JFET accumulation layer under the gate to the drift region. However, the finite-element simulation of Fig. 5(a) tells a different story: the current distribution in the JFET region is highly nonuniform, and current crowding occurs in the region adjacent to the p-base/n-JFET junction depletion layer (see triangular region in Fig. 5(a) for $V_{ds} = 30$ V). Notice that the accumulation layer disappears in the JFET region and is replaced by a depletion region [rectangular region in Fig. 5(a)]. Notice also that the electric potential increases sharply at the end of the channel with drain voltage increasing. A significant portion of the voltage drop is localized in the current spreading region across the depleted portion of the JFET region [see circled region in Fig. 4(a)]. Hence, one can divide the JFET region into two parts: one is the current spreading region across the depleted portion, with electron current spreading out laterally; the other is the triangular region mentioned before, with electron current flowing vertically and nonuniformly. Based on this idea, a new circuit-based model is developed, which physically represents the mechanism of current saturation in power SiC DMOSFET.

The physical treatment of current spreading in DMOSFET proposed by Baliga [17] considers accumulation and JFET region as two individual regions. Resistance of accumulation region is determined by gate voltage only, while resistance of JFET region is a function of drain-source voltage only.

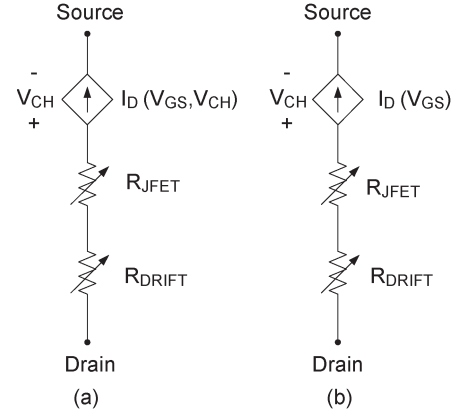


Fig. 6. Standard power MOSFET static model structure.

However, based on the performed finite-element investigation, the resistance of the accumulation region is a strong function of both V_{GS} and V_{DS} , which is not considered in classical solutions for current transport of power DMOSFET. Moreover, the resistance of JFET region is not only determined by depletion width of p-base/JFET junction, but also and more predominantly by reduction of current conduction in the central part of JFET region caused by formation of a depletion region between adjacent p-base regions [rectangular region in Fig. 5(a)]. For large values of gate-source voltage, such as the $V_{gs} = 15$ V case, the JFET current channeling effect is significant only for large values of drain-source voltage, such as the $V_{ds} = 30$ V case of Fig. 5(a), which are not of great practical interest, since power dissipation is exceedingly large. However, for smaller values of gate-source voltage, this effect becomes significant at correspondingly lower values of drain-source voltage. For this reason, it is important to capture this effect in order to have an accurate device model.

III. NOVEL MODEL WITH NONUNIFORM CURRENT DISTRIBUTION IN THE JFET REGION

The simple standard model of a power MOSFET is a piecewise model using different equations to describe operation in the linear region and saturation region. The two equivalent circuits are shown in Fig. 6. For simplicity, device capacitances and the gate terminal are not shown. When channel voltage $V_{CH} < V_{CH,SAT} = V_{GS} - V_T$ (saturation voltage), the MOSFET operates in the linear region, and drain current I_D is dependent on both gate-source voltage V_{GS} and channel voltage V_{CH} . When $V_{CH} \geq V_{CH,SAT}$, the MOSFET operates in the saturation region and I_D is a function of V_{GS} only and does not depend on V_{CH} . Hence, there are two different equations for current I_D in the standard model, one for the linear region and another one for the saturation region. For linear region, the drain current I_D is given by

$$I_D = \frac{\mu_{CH0} C_{OX} Z}{2L_{CH}} [2(V_{GS} - V_T)V_{CH} - V_{CH}^2]. \quad (1)$$

For saturation region, the drain current I_D is

$$I_{DSAT} = \frac{\mu_{CH0} C_{OX} Z}{2L_{CH}} V_{CH,SAT}^2. \quad (2)$$

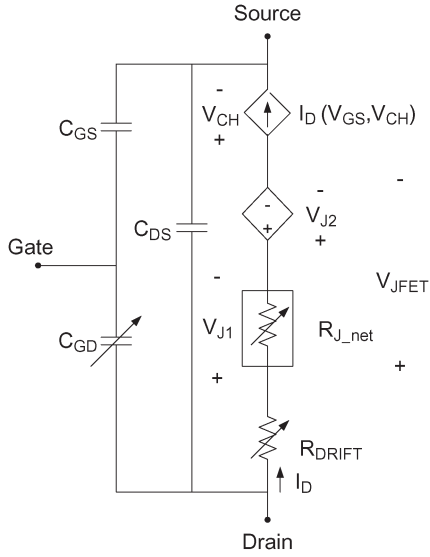


Fig. 7. Proposed SiC DMOSFET model structure.

The nonlinear resistors R_{JFET} and R_{DRIFT} are given by

$$R_{JFET} = \frac{L_{JFET}}{2 \cdot q \cdot \mu_{JFET} \cdot N_{JFET} \cdot Z \cdot (a - W_{JFET})} \quad (3)$$

$$R_{DRIFT} = \frac{L_{DRIFT} - W_{DS}}{2 \cdot q \cdot \mu_{n-drift} \cdot N_{DRIFT} \cdot Z \cdot W_{DRIFT}}. \quad (4)$$

The structure of the new proposed model is shown in Fig. 7. It consists of voltage-controlled current source I_D , voltage source V_{J2} , JFET region resistance net R_{J_net} , drift resistance R_{DRIFT} , and capacitances C_{GS} , C_{DS} , and nonlinear capacitance C_{GD} . The novelty of this model is in how the JFET region is modeled by V_{J2} and R_{J_net} .

Taking into account that SiC MOSFET current saturation is due to a large voltage drop in JFET region and not to channel pinch-off, it is possible to use only one equation to describe the channel region forward I - V characteristic. This is the equation corresponding to linear region of operation in the standard model of Fig. 6

$$I_D = \frac{\mu_{CH} C_{OX} Z}{2 L_{CH}} [2(V_{GS} - V_T)V_{CH} - V_{CH}^2]. \quad (5)$$

Thus, channel region is represented by a voltage-controlled current source, and channel voltage V_{CH} can be determined by subtracting voltage drop in JFET and drift region from total voltage applied to the device

$$V_{CH} = V_{DS} - V_{JFET} - I_D R_{DRIFT} \quad (6)$$

with

$$R_{DRIFT} = \frac{L_{DRIFT} - W_{DS}}{2 \cdot q \cdot \mu_{n-drift} \cdot N_{DRIFT} \cdot Z \cdot W_{DRIFT}} \quad (7)$$

$$W_{DS} = \sqrt{\frac{2\epsilon_{sic}}{q \cdot N_{DRIFT}}} (V_{bi} + V_{CH} + V_{JFET}) \quad (8)$$

where V_{JFET} is the voltage drop in JFET region.

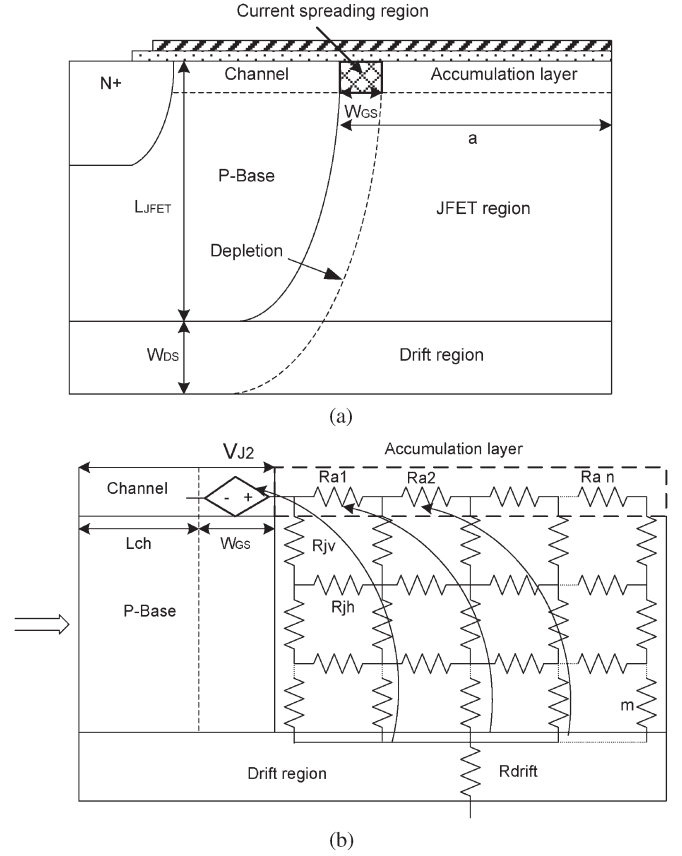


Fig. 8. Model description of JFET region. (a) Specific structure of channel and JFET region. (b) Circuit representation of JFET region.

A. JFET Region Modeling

The proposed method to capture the current saturation in the discussed DMOSFET is to represent JFET region as two parts: a voltage source and a matrix of resistors as in Fig. 7. The specific structure is shown in Fig. 8. Voltage source V_{J2} represents the voltage drop in the current spreading region and the resistor network allows for a nonuniform current distribution in the JFET region. This approach takes into account JFET region voltage drop in both lateral and vertical directions, therefore capturing the 2-D nature of current spreading in this region. The top row of resistors represents the accumulation layer while the remaining matrix resistors represent the main body of JFET region. The main feature of this approach is that values of the resistors in the accumulation layer are function of both V_{GS} and V_{DS} . Let us consider a resistor R_{ai} in the top row. Due to the voltage drop on the resistors to its left, the voltage V_{Gai} between the gate and the resistor node i is decreased, causing reduction of accumulation layer thickness. As a result, it is $R_{a1} < \dots < R_{ai} < \dots < R_{an}$. This effect contributes to current crowding at the depletion layer edge in the JFET region.

To model the voltage drop in vertical direction inside the JFET region, a resistor matrix with n -columns and m -rows is used. First top row of resistors R_{ai} represents accumulation region and can be expressed as

$$R_{ai} = \frac{L_R}{Z \mu_A C_{OX} (V_{GS} - V_T - V_i)} \quad (9)$$

where μ_A is accumulation layer mobility, V_i is voltage at the node of corresponding i resistor, and L_R is the distance between adjacent nodes

$$L_R = \frac{a - W_{GS}}{n} \quad (10)$$

$$W_{GS} = \sqrt{\frac{2\varepsilon_{sic}}{q \cdot N_{JFET}} (V_{bi} + V_{CH} + V_{J2})}. \quad (11)$$

When V_i is smaller than $(V_{GS} - V_T)$, R_i has a finite positive value, representing the accumulation layer resistance. With V_i increasing, the voltage drop $(V_{GS} - V_T - V_i)$ goes to zero and R_i tends to infinity. From a physical point of view, a depletion region forms, replacing the rightmost portion of the accumulation layer [rectangular region in Fig. 5(a)].

JFET region resistive net is represented by horizontal and vertical resistors [see Fig. 8(b)]

$$R_{jh} = \frac{L_R m}{q \mu_{JFET} N_{JFET} Z L_{JFET}} \quad (12)$$

$$R_{jv} = \frac{L_{JFET}}{q \mu_{JFET} N_{JFET} Z L_R m}. \quad (13)$$

The voltage source V_{J2} represents the voltage drop in the current spreading region. It is a critical element in the proposed model, because when the current saturates, this portion supports most of the increased applied voltage. From the finite-element simulation of Fig. 5(b), it can be seen that the electric field reaches its peak value E_k at the end of the channel and decreases approximately linearly to 0 along W_{GS} . Therefore, the amplitude of the voltage source V_{J2} can be obtained by integrating the triangular electric field in the current spreading region, which has a peak value E_k and a length W_{GS}

$$V_{J2} = \frac{1}{2} E_k W_{GS} \quad (14)$$

The peak electric field E_k in the current spreading region can be calculated as follows. The drain current equation at the end of the channel is given by $I_D = C_{OX} \cdot (V_{GS} - V_T - V_{CH}) \cdot Z \cdot v_1$, where v_1 is the electron carrier velocity, which is given by $v_1 = \mu_{CH}(E_k) \cdot E_k$. Combining these two equations, one obtains equation

$$I_D = C_{OX}(V_{GS} - V_T - V_{CH})Z \cdot \mu_{CH}(E_k) \cdot E_k. \quad (15)$$

This implicit equation can be used to calculate peak electric field E_k in the presence of carrier velocity saturation by using (16) to express the field-dependent carrier mobility at the end of the channel

$$\mu_{CH}(E_k) = \mu_{CH0} \cdot \left[1 + \left(\frac{\mu_{CH0} E_k}{v_{sat}} \right)^\beta \right]^{-\frac{1}{\beta}} \quad (16)$$

where μ_{CH0} is the low-field mobility in the inversion layer and v_{sat} is the carrier saturation velocity. Because the peak electric field E_k increases with increasing drain-source voltage, for low voltage, V_{J2} is so small that it does not affect the linear region characteristic. However, for high voltage, V_{J2} supports most of the applied voltage because of the strong electric field due to carrier velocity saturation. As a result, the

channel voltage V_{CH} reaches a constant maximum value, and the device current saturates. As a result, (5) describing channel region conduction is valid under all operating conditions with the nonsaturated mobility $\mu_{CH} = \mu_{CH0}$, because the channel region always operates in the linear operating regime with low electric field. This fact has been verified through finite-element simulations.

B. Nonlinear Capacitance Model

In order to model the dynamic performances of power SiC DMOSFET, nonlinear gate-drain capacitance C_{GD} is introduced in the proposed model shown in Fig. 7. This capacitance is both current and voltage dependent and is comprised of the gate oxide capacitance $C_{OX} A a_i$ in series with the depletion layer capacitance under gate oxide $C_{dep} = \varepsilon_{SiC} A a_i / W'$ [18]. Nonlinear capacitance C_{GD} is given below

$$C_{GD} = \frac{C_{OX} A a_i C_{dep}}{C_{OX} A a_i + C_{dep}} = \frac{C_{OX} A a_i}{1 + C_{OX} \frac{W'}{\varepsilon_{SiC}}} \quad (17)$$

where W' is the length of the depletion region under the gate oxide, a_i is the ratio of intercell area to total die area, C_{OX} is the oxide capacitance per unit area, and A is the total die area

$$W' = W_{DS} - W_{JFET} - a \quad (18)$$

$$W_{JFET} = \sqrt{\frac{2\varepsilon_{sic} V_{GS}}{q \cdot N_{JFET}}} \quad (19)$$

where W_{JFET} is the depletion layer length of the depletion region under the gate, which only begins to grow when $V_{DS} = V_{GS}$, and a is half the width of channel region.

IV. MODEL VALIDATION

The model is implemented in the circuit simulator PSpice and is validated by comparison with Atlas finite-element simulations and experimental measurements. A high-power SiC DMOSFET from CREE Inc., rated at 1200 V 20 A, is used for experimental validation. The device inner construction details are not available from the manufacturer. On the basis of experimental parameter extraction and literature published by CREE [19], the geometrical and doping characteristics of Fig. 1 have been derived. These characteristics are used both for the finite-element simulation model and for the PSpice model. A full set of experimental measurements is performed in order to assist model validation, including: static I - V characteristics, C - V characteristics, and dynamic characterization under resistive switching conditions.

A. Proposed Model Versus Atlas Model

Fig. 9 shows comparison of C - V characteristics between experimental results and finite-element simulated results at $V_{gs} = 0$ V and at measurement frequency $f = 1$ MHz. The finite-element simulation results are accurate for large drain-source voltages, but show discrepancies for lower voltages. Fig. 10 shows comparison of forward characteristics for the proposed new model, standard model, and Atlas model of SiC

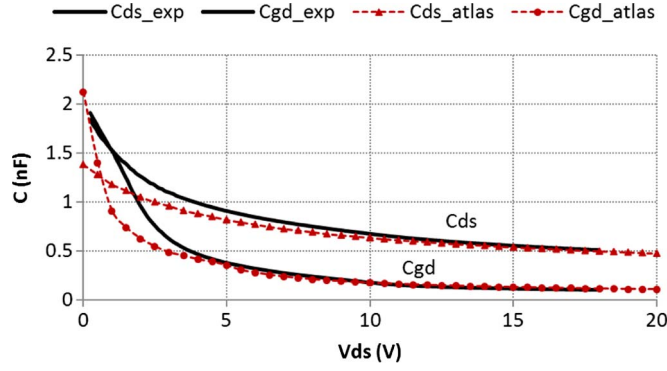


Fig. 9. Comparison of C - V characteristics obtained experimentally and from finite-element simulation.

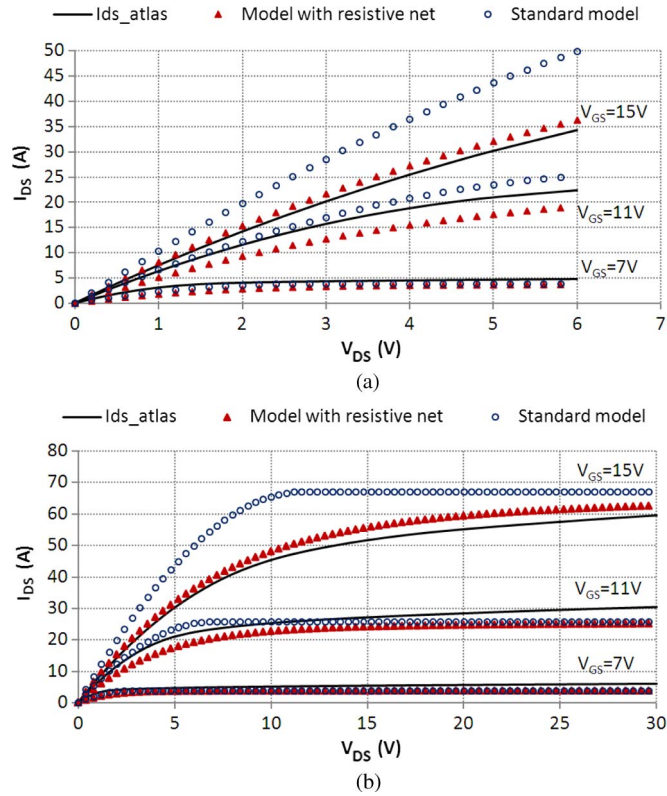


Fig. 10. Comparisons of proposed model, Atlas model, and standard model forward characteristics of SiC DMOSFET for $V_{ds} = 0$ – 7 V (a) and for $V_{ds} = 0$ – 30 V (b).

DMOSFET. Static characteristics obtained using the standard model of Fig. 6 are included for comparison. Fig. 10(a) shows a good match between the proposed model and the Atlas model static characteristics, whereas the standard model predicts significantly larger currents for large gate-source voltages when drain-source voltage V_{DS} is low. For completeness, Fig. 10(b) shows the forward characteristic up to $V_{ds} = 30$ V, showing that for large gate-source voltages ($V_{gs} = 15$ V), the standard model forward characteristic eventually saturates, but it is quite inaccurate in the range $V_{ds} = 3$ – 25 V.

B. Proposed Model Versus Experiment

Static forward characteristics were measured using a Tektronix 371 A power curve tracer for four values of gate-

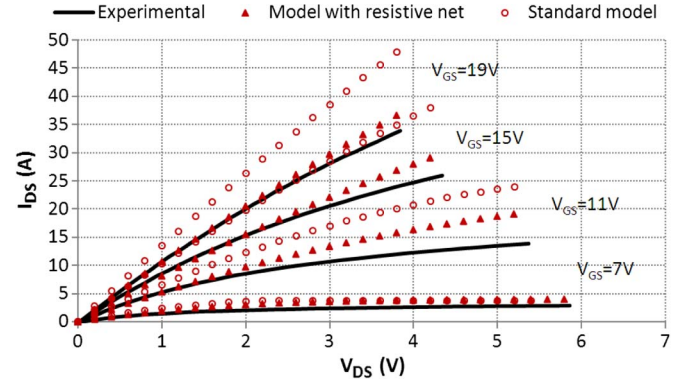


Fig. 11. Comparison of experimental and simulated forward characteristic of SiC DMOSFET.

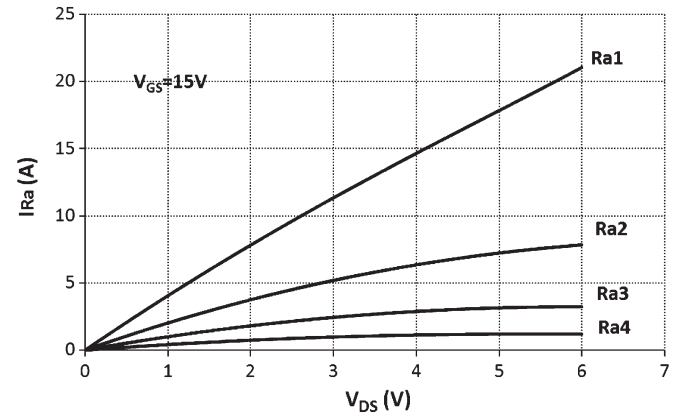


Fig. 12. Model current in accumulation resistors of SiC DMOSFET.

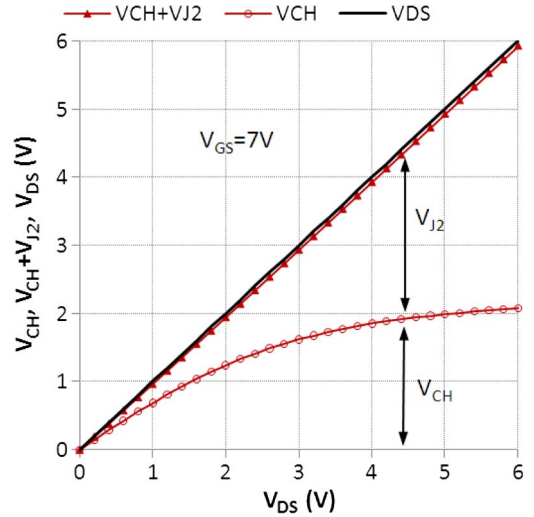


Fig. 13. Channel voltage V_{CH} , voltage source V_{J2} in JFET region and drain-to-source voltage V_{DS} at $V_{gs} = 7$ V.

source voltage: 7 V, 11 V, 15 V, 19 V. The experimental static characteristic of the device is compared with model predictions in Fig. 11. Static characteristics obtained using the standard model of Fig. 6 are included for comparison. Fig. 12 shows simulated result of the current in the individual resistors of the accumulation layer as a function of drain-source voltage (shown for a four-column resistive net). As it can be seen, when drain-source voltage increases, current tends to flow through resistors that are closer to the channel. Fig. 13 shows the channel voltage

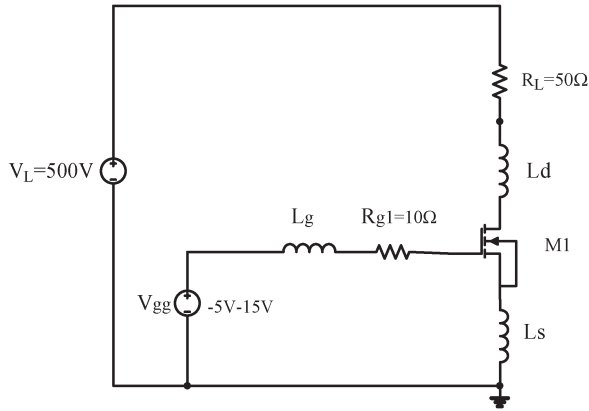


Fig. 14. Equivalent circuit used for resistive switching simulation using the proposed model.

V_{CH} and current spreading region voltage V_{J2} in JFET region at $V_{GS} = 7$ V. The forward characteristic for $V_{GS} = 7$ V shown in Fig. 11 exhibits current saturation as voltage V_{DS} increases. Looking at Fig. 13, one can see that, as drain-source voltage increases, channel voltage tends to saturate while V_{J2} supports most of the voltage drop, and, as a result, device current saturates.

A printed circuit board testbed was built to perform resistive switching experiments on the SiC DMOSFET. Fig. 14 shows the corresponding resistive switching circuit used in simulation, which includes parasitic inductors L_s , L_d , and L_g [20]. Inductor L_d represents the switching loop inductance. Inductor L_s is the MOSFET source-leg parasitic inductance and provides a feedback path from MOSFET drain current to gate-source voltage during transitions. Inductor L_g is the gate circuit loop inductance. Comparison between experimental and simulation results is shown in Fig. 15 for resistive turn-on and in Fig. 16 for resistive turn-off. The simulation results are in good agreement, even if there are some differences, particularly in the waveforms of gate-source voltage V_{GS} . A possible explanation for the discrepancy is that the nonlinear capacitance description in the model has some limitation for small values of drain-source voltages as can be seen in Fig. 9. Further improvements of this capacitance model are left as future work. Figs. 17 and 18 show the turn-on and turn-off waveforms of nonlinear capacitance C_{GD} and the oxide capacitance c_{OX} under resistive switching. These figures show how capacitance C_{GD} varies as a function of drain-source voltage, reaching a maximum value of c_{OX} when drain-source voltage goes to zero.

V. DISCUSSION

The proposed model provides a more physical description of the saturation phenomenon in SiC MOSFETs and is motivated by the presented finite-element simulation results, which show that current saturation is due to carrier velocity saturation at the interface between the channel and the JFET region and to nonuniform current flow in the JFET region. A significant advantage of the proposed model with respect to the conventional MOSFET model is that a single set of equations is valid both in the linear and the saturation region. The conventional piecewise model is unsatisfactory from the point of view of physically

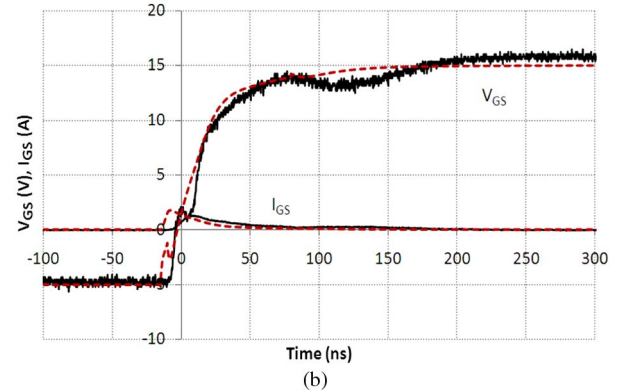
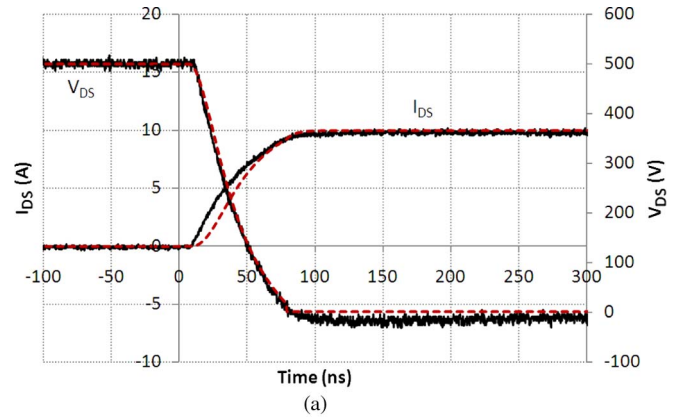


Fig. 15. SiC DMOSFET simulated (dashed) and experimental (solid) turn-on waveforms of resistive switching. (a) Comparison of V_{DS} and I_{DS} ; (b) comparison of V_{GS} and I_{GS} .

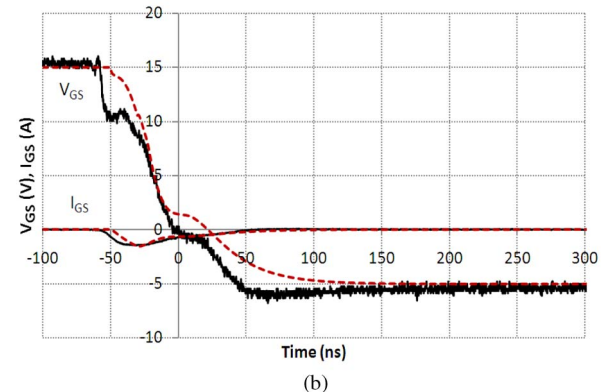
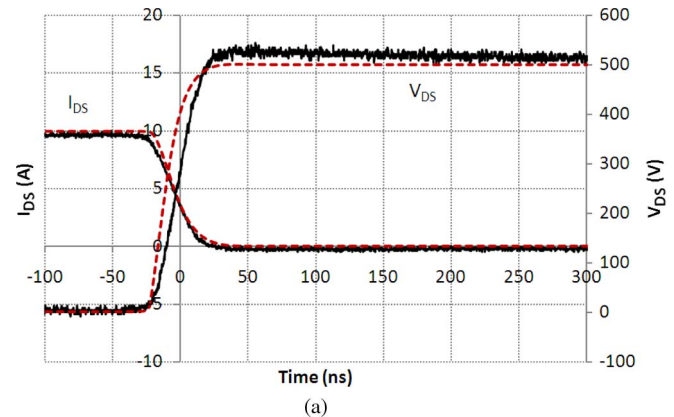


Fig. 16. SiC DMOSFET simulated (dashed) and experimental (solid) turn-off waveforms of resistive switching. (a) Comparison of V_{DS} and I_{DS} . (b) Comparison of V_{GS} and I_{GS} .

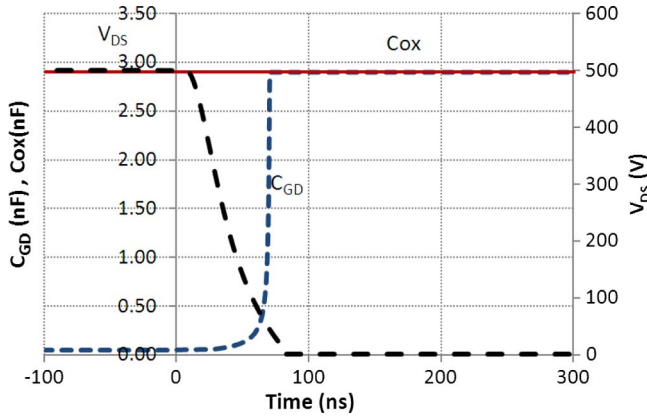


Fig. 17. Time evolution during turn-on of the instantaneous values of non-linear capacitance C_{GD} and of the oxide capacitance c_{OX} under resistive switching.

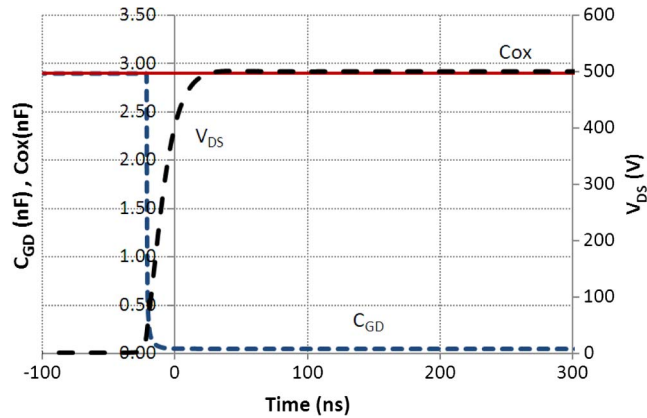


Fig. 18. Time evolution during turn-off of the instantaneous values of non-linear capacitance C_{GD} and of the oxide capacitance c_{OX} under resistive switching.

describing the current saturation phenomenon and is prone to numerical convergence problems when transitioning from one region to the other, due to the jump from one set to another set of describing equations.

The experimental validation shows that the model is capable of accurately describing device operation under both static and dynamic conditions. The actual geometrical dimensions and doping profile for the CREE DMOSFET are not exactly known, as is frequently the case in practice. However, the fact that the static characteristics of the PSpice model match those of the finite-element model indicates that the proposed model captures the physical behavior of the device as described by the finite-element simulation.

This new DMOSFET model provides a more physics-based description of the current saturation phenomenon in state-of-the-art SiC DMOSFETs and can be implemented in circuit-based simulation such as PSpice.

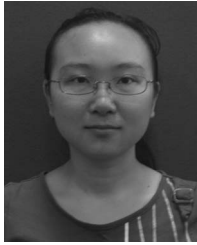
VI. CONCLUSION

A novel DMOSFET model has been proposed. The new feature of the model is that it provides a more physical description of the saturation phenomenon in power MOSFETs, which accounts for JFET region nonuniform current distribution. The

physical basis of the proposed model is motivated by finite-element simulations. The model is validated both statically and under resistive switching conditions for SiC DMOSFET showing overall good matching with experimental results and finite-element simulations.

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